

VALUE ADDED PROBLEM-1 (Experiment – 11)

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/* Name-Simrat Singh Arora
/PRN-25070521181/
SECTION-D-I
Value Added Problem-1 */
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* left;
    struct Node* right;
};
struct Node* createNode(int data) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->left = newNode->right = NULL;
    return newNode;
}
struct Node* insert(struct Node* root, int data) {
    if (root == NULL)
        return createNode(data);
    if (data < root->data)
        root->left = insert(root->left, data);
    else
        root->right = insert(root->right, data);

    return root;
}
void printInRange(struct Node* root, int low, int high) {
    if (root == NULL)
        return;

    if (root->data > low)
        printInRange(root->left, low, high);

    if (root->data >= low && root->data <= high)
        printf("%d ", root->data);

    if (root->data < high)
        printInRange(root->right, low, high);
}
int main() {
    struct Node* root = NULL;
    printf("Name :Simrat Singh Arora\n");
    printf("PRN :25070521181\n");
    printf("Section :D-1\n");
    printf("Value Added Problem : 1\n");
    printf("-----\n");

    root = insert(root, 17);
    insert(root, 4);
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insert(root, 18);
insert(root, 2);
insert(root, 9);
int low = 4, high = 24;
printf("Output: ");
printInRange(root, low, high);

return 0;
}

```

main.c	Run	Output
<pre> 1 /* Name-Simrat Singh Arora 2 /PRN-25070521181/ 3 SECTION-D-I 4 Value Added Problem-1 */ 5 #include <stdio.h> 6 #include <stdlib.h> 7 struct Node { 8 int data; 9 struct Node* left; 10 struct Node* right; 11 }; 12 struct Node* createNode(int data) { 13 struct Node* newNode = (struct Node*)malloc(sizeof(struct 14 Node)); 15 newNode->data = data; 16 newNode->left = newNode->right = NULL; 17 return newNode; 18 } 19 struct Node* insert(struct Node* root, int data) { 20 if (root == NULL) 21 return createNode(data); 22 if (data < root->data) 23 root->left = insert(root->left, data); 24 else 25 root->right = insert(root->right, data); 26 return root; 27 } 28 29 int main() { 30 struct Node* root = NULL; 31 root = insert(root, 18); 32 root = insert(root, 2); 33 root = insert(root, 9); 34 int low = 4, high = 24; 35 printf("Output: "); 36 printInRange(root, low, high); 37 return 0; 38 } </pre>	Run	<pre> Name :Simrat Singh Arora PRN :25070521181 Section :D-1 Value Added Problem : 1 ----- Output: 4 9 17 18 === Code Execution Successful === </pre> Clear